Definition

Study Setup

Data Overview

Data Cleaning and Preprocessing

Exploratory Data Analysis

Research Questions

Analysis of a Contaminated Phase II RCT for Antihypertensive Efficacy

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What is a Contaminated Trial?

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- **Definition:** In clinical research, contamination refers to situations where control group participants receive the intervention or intervention group participants fail to follow the protocol.
- **Impact:** This blurs the distinction between groups, making it harder to detect whether the intervention truly works.
- **Example:** In a trial testing a new health program, if control participants independently adopt parts of the program, the measured effect size will shrink.

Why Contamination Matters?

- **Bias in Results:** Contamination typically leads to an underestimation of treatment effects, as both groups begin to resemble each other.

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- **Reduced Validity:** The internal validity of the trial is compromised, making conclusions less reliable.
- **Policy Implications:** Contaminated trials can mislead healthcare guidelines, especially if the intervention appears less effective than it truly is.

Common Causes of Contamination:

- **Information Sharing:** Participants in different groups communicate and share intervention details.

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- **Healthcare Provider Behavior:** Clinicians may unintentionally provide intervention elements to control patients.
- **Self-Initiated Actions:** Control participants may independently seek out the intervention (e.g., lifestyle changes, medications).
- **Protocol Non-Adherence:** Intervention participants may not fully comply, resembling the control group.

Introduction

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- However, *midway through the trial*, Site 2 switched measurement devices, and there are possible *differential dropout* and suspicions of *protocol deviations* at Site 3.
- The trial statistician has left the organization. We are handed the raw data and told: "Figure it out.".

Objectives

Evaluate whether a new antihypertensive drug (*Trx*) reduces systolic blood pressure (SBP) compared to placebo — BUT the dataset is "contaminated" with multiple real-world issues.

Data Collection and Plan

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- In this case, the dataset was constructed to mimic real-world conditions in healthcare and business.
- This allows us to apply statistical methods as if we were analyzing actual field data, while recognizing the limitations of simulated datasets.

Data Description

- We have got 180 observations with 13 variables where 3 variables are characters, 9 variables are integers and 1 is numerical.
- We got the following variables:
- 1: patient_id: unique patient identifier.
- 2: Site: Clinical site.
- 3: Treatment: "Trx" or "Placebo".
- 4: Age: Age in years (Some are missing).
- 5: Sex: Male or Female.
- 6: Bmi: Body Mass Index (Some are missing).
- 7: sbp_0: systolic BP at baseline (Week 0).
- 8: sbp_4: systolic BP at Week 4.
- 9: sbp_8: systolic BP at Week 8.
- 10: sbp_12: systolic BP at Week 12.
- 11: dropout_week: Week of dropout, NA = completed trial.
- 12: device_flag: = 1 original device, 2 = new device (Site 2 only).
- 13: deviation_flag: 1 = protocol deviation recorded

Scrutiny of Data

After collecting the data we found a lot of information which were incomplete in the disguise of missing.

- Finding the relevant NA Values in "Age" & "Bmi" columns , respectively as the other NA's have specified meaning attached to them.

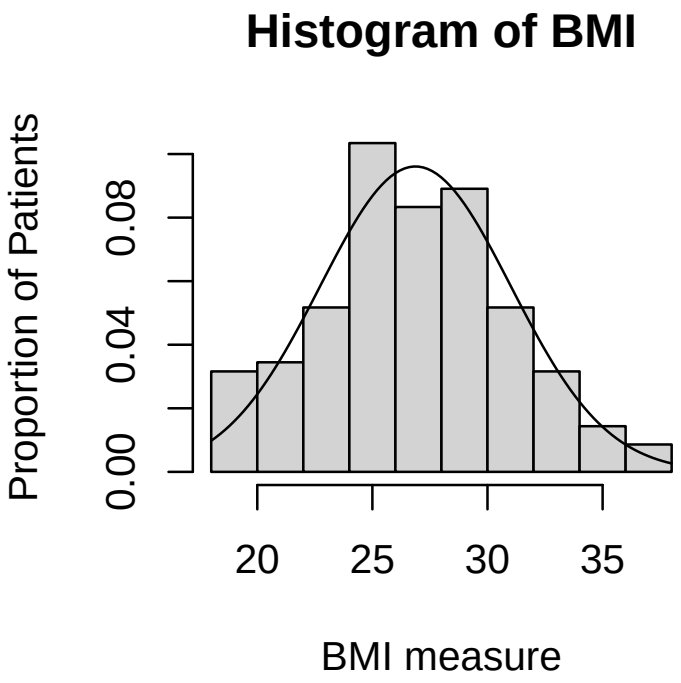
```
    patient_id site treatment age sex  bmi sbp_0 sbp_4 sbp_8 sbp_12
109       P109    2       TrX  NA   M   NA   129   119   127    130
164       P164    3       TrX  NA   F 21.5   158   160    NA     NA
167       P167    3       TrX  NA   M 26.2   127   132   123    119
    dropout_week device_flag deviation_flag drop_flag
109           NA           2               0         0
164            8           1               0         1
167           NA           1               0         0
```

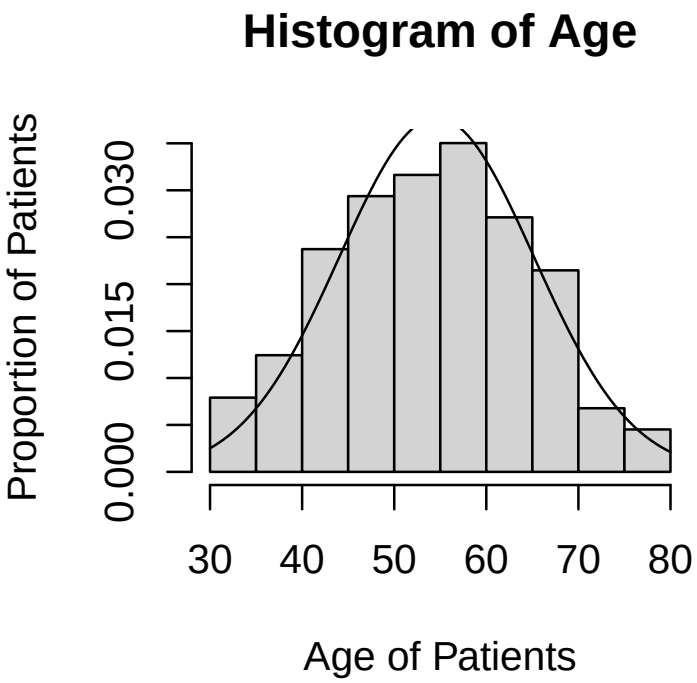
```
    patient_id site treatment age sex bmi sbp_0 sbp_4 sbp_8 sbp_12 dropout_week
52        P052    1   Placebo  54   M  NA   134   130   140    126           NA
70        P070    2       TrX  60   M  NA   153   145   145    138           NA
77        P077    2   Placebo  55   F  NA   145   139   149    139           NA
109       P109    2       TrX  NA   M  NA   129   119   127    130           NA
114       P114    2   Placebo  74   M  NA   139   138   134    129           NA
173       P173    3   Placebo  69   M  NA   153    NA    NA     NA            4
    device_flag deviation_flag drop_flag
52            1              0         0
70            1              0         0
77            1              0         0
109           2              0         0
114           2              0         0
```

Statistical Methodology

Distributional Checking of Key Variables

- We plot the histograms of "BMI" and "Age" variables to see whether they follow normal distribution or not!?





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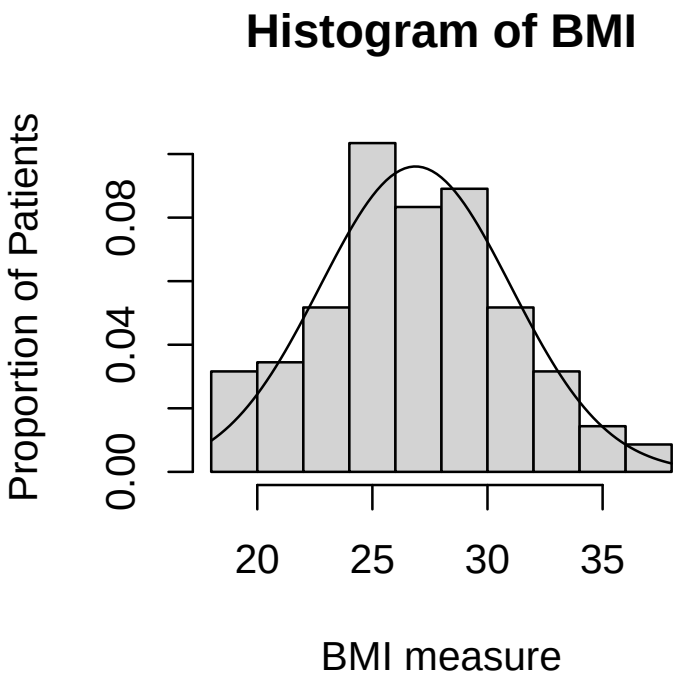

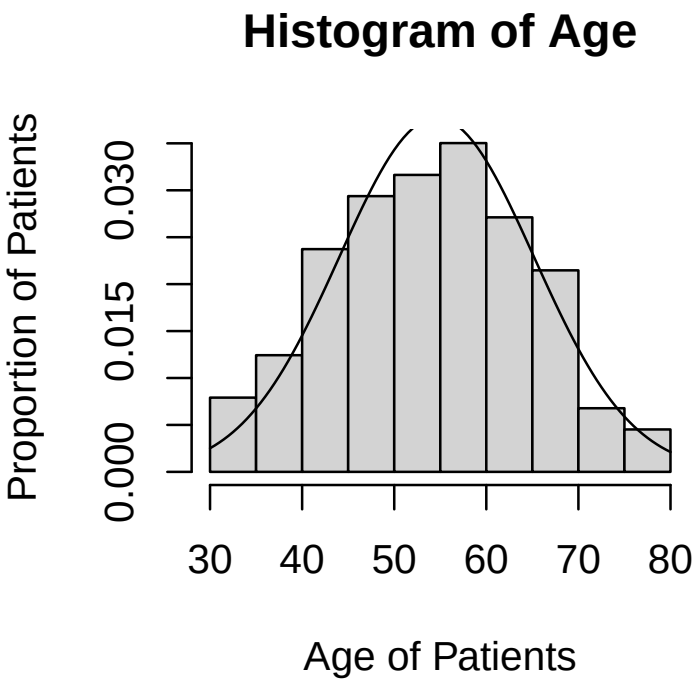

- It seems that both the variables follow normal distribution.

Statistical Methodology (Contd.)

Distributional Checking of Key Variables

- We can check it via formal testing also. We shall use Shapiro Wilk's test for checking normality assumption.

```
 Shapiro-Wilk normality test

data:  dat$bmi
W = 0.99268, p-value = 0.5294

 Shapiro-Wilk normality test

data:  dat$age
W = 0.99183, p-value = 0.4172
```

Statistical Methodology (Contd.)

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- Very large p-value (>> 0.05) and hence no evidence against H_0.

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```

- Very large p-value (>> 0.05) and hence no evidence against H_0.
- We found that the assumption for normality is valid.

Variable Pre-processing

- We now replace the NA values in "Age" & "Bmi" by generating values from truncated normal distribution, as the "Age" & "Bmi" follows normal distribution.

Re-checking NA Values


```
 [1] patient_id     site           treatment      age            sex
 [6] bmi            sbp_0          sbp_4          sbp_8          sbp_12
[11] dropout_week   device_flag    deviation_flag drop_flag
<0 rows> (or 0-length row.names)
```

```
 [1] patient_id     site           treatment      age            sex
 [6] bmi            sbp_0          sbp_4          sbp_8          sbp_12
[11] dropout_week   device_flag    deviation_flag drop_flag
<0 rows> (or 0-length row.names)
```

- No "NA" values recorded.

Exploratory Data Analysis

Data Quality Report

- Nextly, Outlier detection - we choose and justify our method and flag any identified outliers explicitly.

```
Outliers in bmi :
[1] 37.7

Outliers in sbp_8 :
[1] 100 197

Outliers in device_flag :
 [1] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2

Outliers in deviation_flag :
 [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

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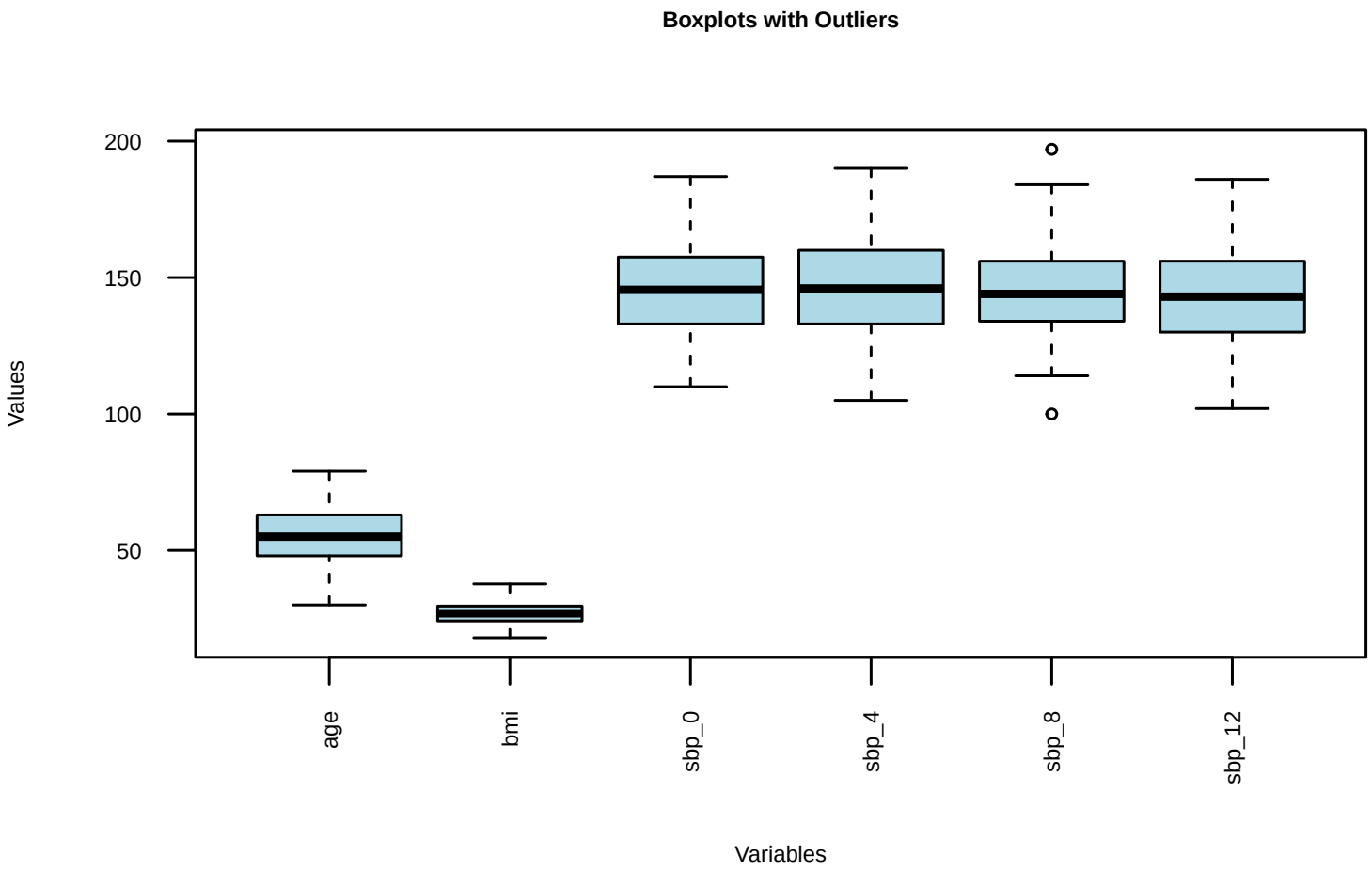
Outliers in deviation_flag :
 [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

- Outliers were identified using Tukey's IQR method. All total outliers were observed in continuous variables such as SBP and BMI. These may reflect either genuine clinical variability or potential measurement inconsistencies.

Exploratory Data Analysis (Contd.)

Data Quality Report

- We can visualize the outliers as follows via boxplot.



Exploratory Data Analysis (Contd.)

Data Quality Report

- We are interested in checking the Balance of baseline covariates (age, sex, BMI) across treatment arms.
- Following is the Table to check balance of baseline covariates.

```
                 Stratified by treatment
                  level Placebo       TrX           p      test
n                          90            90
age (mean (SD))        54.46 (9.76)  54.86 (11.38)  0.800
sex (%)           F        46 (51.1)     41 (45.6)   0.551
                  M        44 (48.9)     49 (54.4)
bmi (mean (SD))        26.96 (4.05)  26.80 (4.26)   0.803
```

- We notice that Baseline covariates including age, sex, BMI were well balanced between the treatment and placebo groups (all p-values $>$ 0.05), indicating no class imbalance.

Exploratory Data Analysis (Contd.)

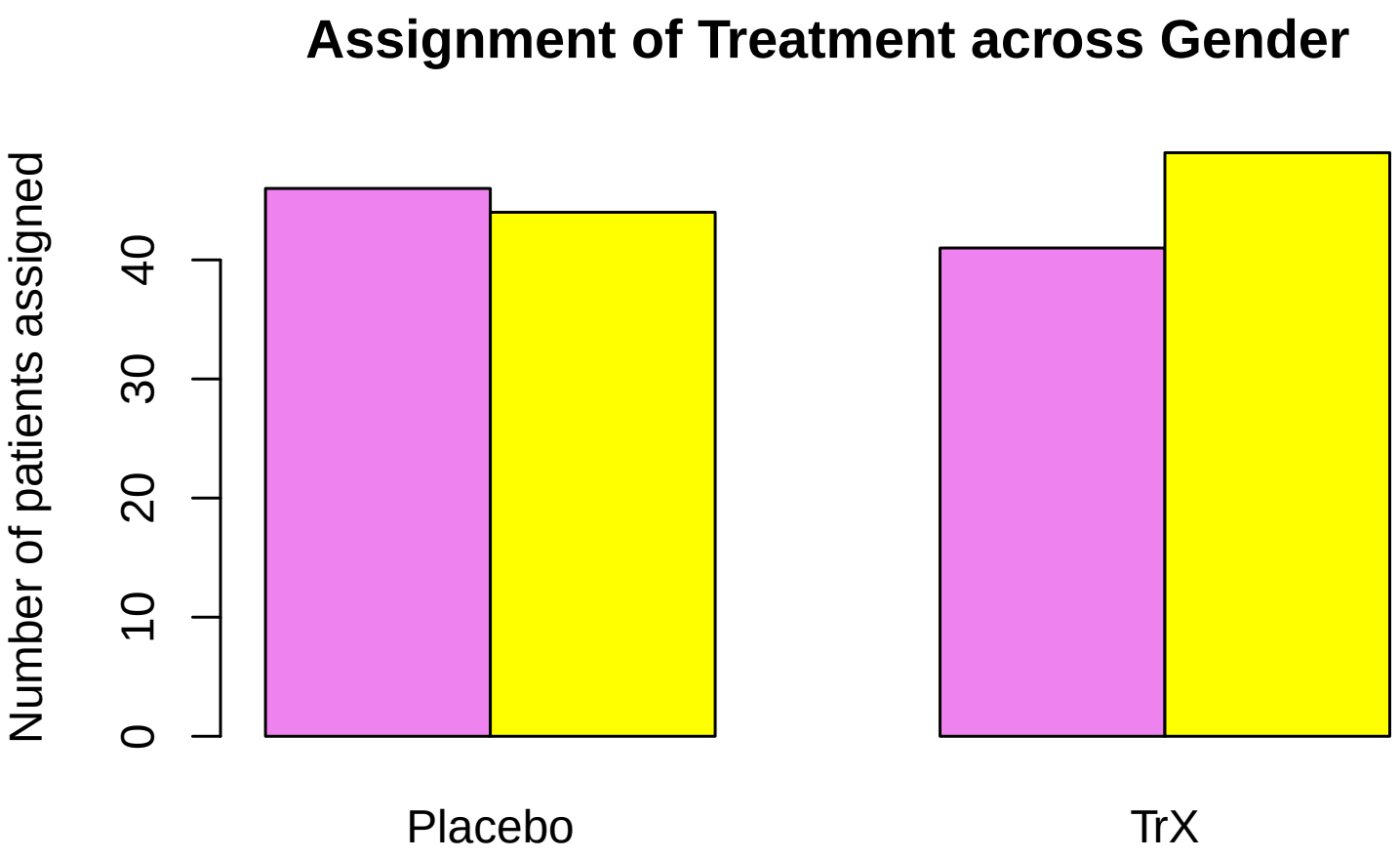

```
   0  1
1 60  0
2 60  0
3 45 15
```

- We checked and results showed that protocol deviation recorded at site 3. We might assess it later.

Exploratory Data Analysis (Contd.)

Data Quality Report

- Nextly, we can visualize the assignment of treatment across gender.



Gender (Violet−Female, Yellow−Male)

Exploratory Data Analysis (Contd.)

Data Quality Report

- We tested whether sex and treatment assignment were independent or not.

```
    Placebo TrX
  F      46  41
  M      44  49

 Pearson's Chi-squared test with Yates' continuity correction

data:  tab
X-squared = 0.35595, df = 1, p-value = 0.5508
```

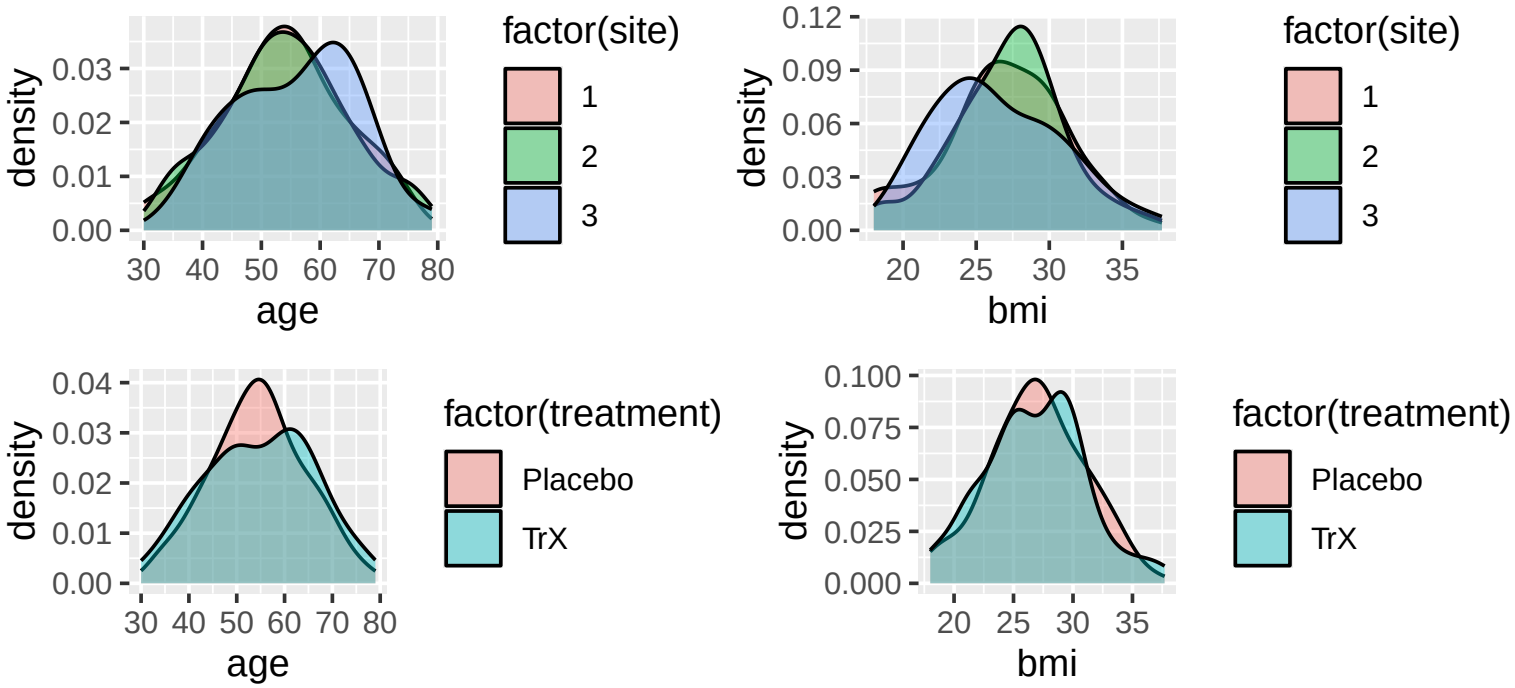
- The result indicates no significant association. Thus, the treatment groups appear balanced with respect to sex.

Exploratory Data Analysis (Contd.)

Distributional Checking

- We should take into account Distributional checks per site and treatment arm (visual and/or formal).

```
Attaching package:  'patchwork'
The following object is masked from 'package:MASS':
    area
```



Exploratory Data Analysis (Contd.)

Distributional Checking

- We notice that all sites roughly normally distributed.
- Although, Site 2 much more centered and narrow maybe due to device change,only suspect no conclusive evidence. Needed to investigate later.

Research Questions

- Does Trx reduces SBP compared to placebo?
- Did the device change at Site-2 affect measurements?
- Is the dropout systematic or random (MAR)?
- Is treatment effect consistent across site?
- Are results robust to contaminations (Site-3 issues)?

Inferential Analysis

Treatment Effect Definition

- We denote our primary endpoint is change in SBP from baseline to Week 12, defined as: *delta_sbp = sbp_0 - sbp_12.*

```
dat$del_sbp=(dat$sbp_0-dat$sbp_12)
```

- So, we checking normality for the variable *delta_sbp.*

```
 Shapiro-Wilk normality test

data:  dat$del_sbp
W = 0.98423, p-value = 0.1517
```

- The test suggests that it normally distributed.

Inferential Analysis (Contd.)

Model Specification

- A linear regression model was specified to evaluate the effect of treatment on change in sbp, defined as the difference between baseline and Week 12 values.
- The model adjusted for baseline SBP, demographic variables (age, sex, BMI), and study site to control for potential confounding and site-level heterogeneity.

```
t test of coefficients:

               Estimate Std. Error t value  Pr(>|t|)
(Intercept)  -9.5059332  8.0072003 -1.1872    0.2375
sexM          0.6928261  1.2388155  0.5593    0.5770
site2        -2.1717085  1.3240736 -1.6402    0.1036
site3        -1.8122828  1.9711961 -0.9194    0.3598
age           0.0944434  0.0589304  1.6026    0.1117
bmi           0.1300915  0.1721197  0.7558    0.4513
treatmentTrX  8.7986819  1.2045636  7.3045 3.564e-11 ***
sbp_0         0.0061326  0.0383543  0.1599    0.8732
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- The treatment (Trx) showed a statistically significant effect.

Inferential Analysis (Contd.)

Device Switch at Site - 2

- Now we are deeply look into Formally test whether the device switch at Site 2 introduced a systematic measurement shift or not.
- Divide the SBP across 12 weeks into two segments, before and after the device change,specific to Site-2 (Clear off NA values).

```
a=(dat[which(dat$site==2),]$sbp_0+dat[which(dat$site==2),]$sbp_4)/2;a=na.omit(a)
b=(dat[which(dat$site==2),]$sbp_8+dat[which(dat$site==2),]$sbp_12)/2;b=na.omit(b)
```

- Then test the Normality assumptions for both groups.

```
shapiro.test(a);shapiro.test(b)


 Shapiro-Wilk normality test

data:  a
W = 0.98277, p-value = 0.5905

 Shapiro-Wilk normality test

data:  b
W = 0.97433, p-value = 0.2858
```

- Normality assumptions are valid.

Inferential Analysis (Contd.)

Device Switch at Site - 2

- So we test the equal variance assumption of the two groups.

```
 F test to compare two variances

data:  a and b
F = 0.84472, num df = 56, denom df = 54, p-value = 0.5324
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.4944876 1.4392335
sample estimates:
ratio of variances
         0.8447197
```

- As p-value > 0.05, we accept the null and hence, the equal variance assumption is clearly valid.
- Our main focus is to test of mean for two groups to notice any Statistically Significant Difference in Observation.

```
 Two Sample t-test

data:  a and b
t = 0.72685, df = 110, p-value = 0.4689
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -3.807437  8.217963
sample estimates:
mean of x mean of y
 148.1053  145.9000
```

- The results showed no statistically significant difference between the original and new device.

Inferential Analysis (Contd.)

Dropout Mechanism

- We try to Assess whether dropout is Missing At Random (MAR) or not.
- We simply do logistic regression to check if dropout is MAR.

```
Call:
glm(formula = drop_flag ~ age + sex + bmi + sbp_0 + treatment +
    site, family = binomial, data = dat)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -6.808312   3.446374  -1.975   0.0482 *
age           0.038479   0.023539   1.635   0.1021
sexM          0.142429   0.488367   0.292   0.7706
bmi           0.028514   0.058466   0.488   0.6258
sbp_0         0.007218   0.015167   0.476   0.6341
treatmentTrX -0.455090   0.482144  -0.944   0.3452
site2         0.548951   0.760331   0.722   0.4703
site3         4.259601   0.697304   6.109    1e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 219.91  on 179  degrees of freedom
Residual deviance: 119.34  on 172  degrees of freedom
AIC: 135.34
```

Inferential Analysis (Contd.)

Dropout Mechanism


```
(Intercept)          age         sexM          bmi        sbp_0 treatmentTrX
      0.001        1.039        1.153        1.029        1.007        0.634
      site2        site3
      1.731       70.782
```

- Dropout was found to be strongly associated with study site, with participants at Site 3 having significantly higher odds of dropout ($p < 0.001$).

Inferential Analysis (Contd.)

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- Age showed a marginal association, while other variables were not significant predictors.

Inferential Analysis (Contd.)

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```

- Dropout was found to be strongly associated with study site, with participants at Site 3 having significantly higher odds of dropout ($p < 0.001$).
- Age showed a marginal association, while other variables were not significant predictors.
- Since dropout is associated with observed variables (particularly site), the missingness mechanism is consistent with Missing At Random (MAR).

Inferential Analysis (Contd.)

Site-level Analysis with Multiplicity Correction

- Here we Test the same primary hypothesis separately within each site.
- Multiplicity correction was applied to account for multiple hypothesis testing in site-specific analyses, reducing the risk of false-positive findings. Adjusted p-values were obtained using the *Bonferroni* method. (why? As it's Very conservative. Reduces false positives strongly).

```
[1] 9.271406e-06 1.448223e-04 4.219813e-01
```

- Site-wise analyses showed that the treatment effect was statistically significant in Site 1 and Site 2 even after Bonferroni correction, while no significant effect was observed in Site 3. This partially agrees with the overall analysis, where a strong treatment effect was detected. The lack of significance in Site 3 may be attributed to high dropout rates and protocol deviations. Overall, the treatment effect appears robust but exhibits heterogeneity across sites.

Inferential Analysis (Contd.)

Primary Hypothesis Test

- Nextly our main objective to Test the primary null hypothesis: Trx produces no mean reduction in SBP compared to Placebo at Week 12.
- We are using t-test of Treatment coefficient to check null hypothesis

```
         Estimate Robust_SE  t_value  df p_value CI_lower CI_upper
Estimate 8.798682  1.204564 7.304456 118       0 6.413318 11.18405
         Effect_Size_d
Estimate      1.324487
```

- The treatment effect was highly significant, indicating that the treatment group experienced a substantially greater reduction in SBP compared to placebo.

Robustness and Sensitivity

Sensitivity Analysis Excluding Site 3

- We are interested to Conduct a sensitivity analysis by excluding Site 3 entirely due to protocol deviations.

```
Call:
lm(formula = del_sbp ~ treatment + sbp_0 + age + sex + bmi +
    factor(site), data = df_sens)

Residuals:
     Min       1Q   Median       3Q      Max
-14.5810  -5.1295   0.3868   4.0604  23.1405

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   -8.805317   8.198154  -1.074   0.2853
treatmentTrX   9.027459   1.287230   7.013 2.35e-10 ***
sbp_0          0.006441   0.040948   0.157   0.8753
age            0.090864   0.061933   1.467   0.1453
sexM           0.595404   1.313720   0.453   0.6513
bmi            0.106843   0.159931   0.668   0.5056
factor(site)2 -2.158656   1.275984  -1.692   0.0937 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.728 on 105 degrees of freedom
  (8 observations deleted due to missingness)
Multiple R-squared:  0.3564, Adjusted R-squared:  0.3196
F-statistic: 9.689 on 6 and 105 DF,  p-value: 1.779e-08
```

Robustness and Sensitivity (Contd.)

Sensitivity Analysis Excluding Site 3

- We compare the original model coefficients and the new model coefficients which excludes site 3.

```
abs_change

treatmentTrX
   0.2287772

perc_change

treatmentTrX
     2.60013
```

Robustness and Sensitivity (Contd.)

Sensitivity Analysis Excluding Site 3

- A sensitivity analysis was conducted by excluding Site 3 due to protocol deviations and high dropout. The treatment effect remained positive and highly statistically significant (Estimate = 9.027, $p < 0.001$), indicating that the direction and significance of the primary conclusion did not change after removing Site 3.
- The original treatment effect estimate was approximately 8.79, while the sensitivity analysis yielded 9.027, resulting in an absolute change of 0.237 mmHg and a percentage change of 2.69%.

Final Conclusions

Key Findings

- This shift is minimal, indicating that the treatment effect estimate is stable and not materially influenced by Site 3.
- Overall, these findings demonstrate that the results are robust to the exclusion of Site 3.
- Despite earlier concerns regarding high dropout and potential bias at this site, its removal does not meaningfully alter the estimated treatment effect. This suggests that the primary findings are not driven by site-specific anomalies.

Final Conclusions (Contd.)

Limitations

- From a trial credibility perspective, this strengthens confidence in the results. The consistency of the treatment effect across analyses indicates that the conclusions are reliable and generalizable, and that the observed efficacy of the treatment is not sensitive to data irregularities at a single site. However, the observed heterogeneity and dropout issues at Site 3 should still be acknowledged as a limitation in interpretation.

THANK YOU!